

DISEASE PREDICTION USING MACHINE LEARNING

CodeAlpha Internship Project

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Internship Domain:

Machine Learning

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CodeAlpha

DISEASE PREDICTION USING MACHINE LEARNING

1. Introduction

Disease prediction is an important application of Machine Learning in the healthcare sector. By analyzing patient medical data, machine learning algorithms can identify patterns and assist in predicting diseases at an early stage. Early disease detection helps healthcare professionals make informed decisions and improve patient outcomes. In this project, the Breast Cancer Wisconsin Dataset is used to build and evaluate machine learning models for disease prediction.

2. Objective

The main objective of this project is to develop a machine learning model capable of predicting disease outcomes based on medical attributes. The project also aims to compare the performance of different classification algorithms and identify the most accurate model.

3. Dataset Description

The Breast Cancer Wisconsin Dataset is a well-known healthcare dataset used for classification tasks.

Dataset Details:

- Total Records: 569
- Features: 30 medical attributes
- Target Variable: Disease Classification (Benign or Malignant)
- Dataset Source: Scikit-learn Built-in Dataset

The dataset contains information about cell nuclei obtained from breast mass images and is commonly used for machine learning research.

4. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

5. Methodology

Step 1: Data Collection

The Breast Cancer Wisconsin Dataset was loaded using the Scikit-learn library.

Step 2: Data Preprocessing

The dataset was converted into a Pandas DataFrame. Missing values and dataset structure were analyzed before training the model.

Step 3: Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Step 4: Model Training

Two machine learning algorithms were used:

1. Logistic Regression
2. Random Forest Classifier

Step 5: Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

6. Results

Logistic Regression

- Accuracy: 95.61%

Random Forest Classifier

- Accuracy: 96.49%

Classification Report

Metric	Value
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Precision	96%
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Metric	Value
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Recall	95%
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F1-Score	95%
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Confusion Matrix

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[[39  4]
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 [ 1 70]]
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The Random Forest model achieved the highest accuracy and demonstrated excellent prediction performance.

7. Conclusion

This project successfully implemented a Disease Prediction System using Machine Learning techniques. Both Logistic Regression and Random Forest Classifier produced highly accurate predictions. Among the two models, Random Forest achieved the best performance with an accuracy of 96.49%. The project demonstrates how machine learning can be effectively applied in healthcare for disease prediction and decision support.

8. Future Enhancements

- Use larger healthcare datasets for improved prediction accuracy.
- Apply advanced machine learning and deep learning algorithms.
- Develop a web-based application for disease prediction.
- Integrate real-time patient data for practical healthcare usage.